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# Run3 performance check with photon-involved channels

Yingrui Hou[1], Maximilien Chefdeville[1].

[1]*LAPP, CNRS-IN2P3, Annecy, France*

**Abstract**

In this document, data performance checks with photon-involved channels are performed. Two channels with high-PT photon, $B_s \to \phi\gamma$ and $B_d \to K^*\gamma$, and two with low-PT photon/$\pi^0$, $B^+ \to \chi_{cJ}(\to J/\psi\gamma)K^+$ and $B^+ \to J/\psi K^{*+}(\to K^+\pi^0)$, are used as benchmark channels to check the performance both in Run3 MC and data (2022 and 2023 data).

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# 1 Introduction

Reconstruction and identification of photons and neutral pions at LHCb is the most important factor in radiative decays, some open-charm and charmonium B meson decay analysis. These particles are reconstructed as neutral clusters by the tracking and calorimeter system. The photon and neutral pion reconstruction performance (both reconstruction efficiency and identification) were studied carefully in Run1 and Run2 with the calibration channels and samples (with TURCAL lines), making it meaningful and reasonable to do similar checks with early Run3 data.

Except for the motivation described in the previous paragraph, there are two other main reasons for the performance check with early Run3 data:

- The pre-shower and SPD detectors are removed in Run3 era [1], and these two detectors were used to be used as part of the identification system, and a set of related parameters were assigned in Run1 and Run2 to align and calibrate these two detectors. So, it's quite necessary to check the photon reconstruction performance both in low pT region and high pT region with Run3 data and MC to validate the ECAL calibration and alignment.

- The hardware trigger (L0 trigger) is removed in Run3 era, instead, the fully-software real-time trigger, alignment and calibration framework is deployed [1]. Since Calo, especially ECAL, is used to be the main L0 trigger detector, it's interesting and meaningful to check the photon reconstructions without L0 trigger and also useful to check the distributions of photons under the new Run3 software framework as validation.

## 1.1 Samples

The data sample used in this document is from 2022 commissioning data processed earlier this year, and 2023 commissioning, and collision data from Turbo stream and Spruce stream. The details about the samples are shown in the Tab. 1.

## 1.2 Hlt2 and spruce lines

Since the two charmonium lines are not included in the Moore version that was used in the 2022 data-taking period, and there's no spruce line for these two decay channels, the checks are only done with Collision23 data(HLT2) and Commissioning23 data. For the two radiative decays, the Hlt2 exclusive lines, Hlt2 inclusive line, and spruce exclusive lines are developed for a variety of analysis purposes. For the full sprucing stream, the event will be filtered firstly by the Hlt2 inclusive line `Hlt2RD_BToHHGamma`, and then the spruce exclusive line `SpruceRD_BdToKstGamma` and `SpruceRD_BsToPhiGamma`. For the turbo sprucing stream, the event will be filtered directly by the Hlt2 lines (`Hlt2RD_BdToKstGamma, Hlt2RD_BsToPhiGamma, Hlt2B2CC_BuToChicKPlus, Hlt2B2CC_BuToJpsiKstPlus`). The details of the selections in lines are shown in Tab. 2 and Tab. 3. [1] [2]

[1] The mva model used in inclusive Hlt2 line is based on the similar model built for run2,variables

[2] The m(hh) and PID selections of $B^0 \to K^*\gamma$ are slightly looser in 2022 data taking time for validating, to make the trigger rate at a reasonable level, the prescale factor is set to be 0.75. This prescale factor is removed in lines used in 2023 data taking.

| | |
|---|---|
| Collision22 data | BKK: /LHCb/Collision22/Beam6800GeV-VeloClosed-MagUp(Down)-Excl-UT/Real Data/94000000/RAW<br>Detectors: Velo closed, without UT<br>Software: From Moore v54r2 - Moore v54r3p1; DaVinci v63r6<br>Alignment: `AlignmentV9-2023-03-16-VPSciFiRich`<br>Early measurement run lumi: 19.7 $pb^{-1}$ from online lumi and offline data validation[a] |
| Commissioning23 data | BKK: /LHCb/Commissioning23/Beam6800GeV-VeloClosed-MagDown-Excl-UT/Real Data/SprucingPass23/94000000/RD(B2CC).DST<br>Dectecots: Velo gap (47mm, half:23.5mm), without UT<br>Software: Moore v54r15; DaVinci v63r13p1<br>Alignment: `Alignment2023-HLT2Processing-26-8-2023`<br>Luminosity: 34 $pb^{-1}$ |
| Collision23 data(HLT2) | BKK: /LHCb/Collision23/Beam6800GeV-VeloClosed-MagDown-Excl-UT/Real Data/Sprucing23r1/90000000/RD(B2CC).DST<br>Detectors: Velo gap (47mm, half:23.5mm), without UT<br>Software: Moore v54r18; DaVinci v63r13p1<br>Alignment: `Alignment2023-HLT2Processing-26-8-2023`<br>Luminosity: 47 $pb^{-1}$ |
| Collision23 data(Spruce) | BKK: /LHCb/Collision23/Beam6800GeV-VeloClosed-MagDown-Excl-UT/Real Data/Sprucing23r1/90000000/RD.DST<br>Detectors: Velo gap (47mm, half:23.5mm), without UT<br>Software: Moore v54r19; DaVinci v63r13p1<br>Alignment:`Alignment2023-HLT2Processing-26-8-2023`<br>Luminosity: 47 $pb^{-1}$ |

Table 1: Details of 2022, 2023 data samples, both turbo spruce and fullstream spruce.

[a]https://twiki.cern.ch/twiki/bin/viewauth/LHCbInternal/CommissioningData2022

| Vars | `Hlt2RD_BsToPhiGamma` | `Hlt2RD_BdToKstGamma` | `Hlt2RD_BToHHGamma` | `SpruceRD_BsToPhiGamma` | `SpruceRD_BdToKstGamma` |
|---|---|---|---|---|---|
| Track $p$ | >2 GeV | >3 ($\pi$: >2) GeV | - | >2 GeV | >3 ($\pi$: >2) GeV |
| Track $pT$ | >0.5 GeV | >0.5 GeV | [1.0, 25] GeV | >0.5 GeV | >0.5 GeV |
| Track IPCHI2 | >11 | > 11 | > 7.4 | > 11 | > 11 |
| track PID | $\mathrm{K_{PIDK}} > 0$ | $\mathrm{K_{PIDK}} > 0, \mathrm{K_{PIDK}} - \mathrm{K_{PIDP}} > 2$<br>2022: $\mathrm{K_{PIDK}} > 0$ | - | $\mathrm{K_{PIDK}} > 0$ | $\mathrm{K_{PIDK}} > 0$ |
| $p_{\mathrm{hh}}$ | >1.5 GeV | >1.5 GeV | >2 GeV | >1.5 GeV | >1.5 GeV |
| Vertx chi2ndof(hh) | <15 | <15 | - | <15 | <15 |
| mass(hh) | [0.95, 1.1] GeV | [0.8, 1.0] GeV | - | [0.95, 1.1] GeV | [0.705, 1.085] GeV |
| photon ET | >2 GeV | > 2 GeV | > 2 GeV | > 2 GeV | > 2 GeV |
| isPhoton | >0.4 | >0.4 | - | >0.4 | >0.4 |
| isNotH | >0.3 | >0.3 | - | >0.3 | >0.3 |
| E19 | >0.2 | >0.2 | - | >0.2 | >0.2 |
| mass(B) | [4.2, 6.4] GeV | [4.2, 6.4] GeV | - | [4.2, 6.4] GeV | [4.2, 6.4] GeV |
| IPCHI2(B) | <12 | < 12 | - | < 15 | < 15 |
| dira(B) | $> \cos(0.1)$ | $> \cos(0.045$ | > 0 | $> \cos(0.1)$ | $> \cos(0.045$ |
| mva(B) | - | - | > 0.22 | - | - |
| prescale | 1.0 | 1.0 (2022: 0.75) | 1.0 | 1.0 | 1.0 |

Table 2: Selections in Hlt2 and spruce lines of radiative channels.

| Vars | Hlt2B2CC_BuToJpsiKstPlus | Hlt2B2CC_BuToChicKPlus |
|---|---|---|
| K pt | $> 0.5$ GeV | |
| K p | $> 2$ GeV | |
| K chi2ndof | $< 10$ | |
| K PID | $K_{PIDK} > 0$ | |
| $\mu$ pt | $> 0.5$ GeV | |
| $\mu$ p | $> 0$ GeV | |
| $\mu$ chi2ndof | $< 5$ | |
| $\mu$ IPCHI2 | $> 0$ | |
| $\mu$ PID | PIDMU>-5. ISMUON=True | |
| $J/\psi$ mass | [2.95, 3.25] GeV | |
| $J/\psi$ chi2ndof | $< 20$ | |
| $J/\psi$ pt | $> 0.5$ GeV | |
| $J/\psi$ docachi2 | $< 20$ | |
| $\gamma(\pi^0)$ pt | $> 1.0$ GeV | $> 0.5$ GeV |
| $\chi_{cJ}(K^{*+})$ mass | [3.41, 3.61] GeV | [0.792, 0.992] GeV |
| $\chi_{cJ}(K^{*+})$ pt | $> 1.0$ GeV | $> 1.5$ GeV |
| B mass | [4.8, 6.1] GeV | |
| B IP | $< 0.2$ | |
| B IPCHI2 | $< 20$ | |
| B dira | $> 0.9995$ | |
| B vertex chi2ndof | $< 10$ | |

Table 3: Selection details in Hlt2 lines of two charmonium channels

## 1.3 Tuple production

All the data tuples mentioned in this document are produced by the `AnalysisProduction`, they can be easily found under the RD and B2CC labels. The locations of the productions are summarized in Tab. 4 and the related merge request and DaVinci codes can be accessed from the production page.

| Production | AP locations (link) |
|---|---|
| Radiative 2022 data | MagUp |
| | MagDown |
| Radiative 2023 turbo data | Commissiong23 |
| | Collision23 |
| Radiative 2023 spruce data | Collision23 |
| Charmonium 2023 turbo data | Commisionning23 |
| | Collision23 |

Table 4: Locations of the tuple productions that are used in this note.

# 2 Radiative channels

For the radiative channels, some offline selections are performed, based on the Run1 early measurement, summarized in the Tab. 5 To check and validate the offline selections,

| Var | $B^0 \to K^*\gamma$ | $B_s \to \phi\gamma$ |
|---|---|---|
| m(hh)-m(PDG) | $< 50$ MeV | $< 10$ MeV |
| hh open angle | $> 0.0004$ | $> 0.0004$ |
| hh vertex chi2 | $< 9$ | $< 9$ |
| photon ET | $> 2.6$ GeV | $> 2.6$ GeV |
| B pT | $> 3$ GeV | $> 3$ GeV |
| B IPCHI2 | $< 9$ | $< 9$ |
| B cos(helicity)\| | $< 0.8$ | $< 0.8$ |
| B dira | $> 0.999$ | $> 0.999$ |
| $\epsilon_{\rm rec}$ | 2.1% | 2.5% |

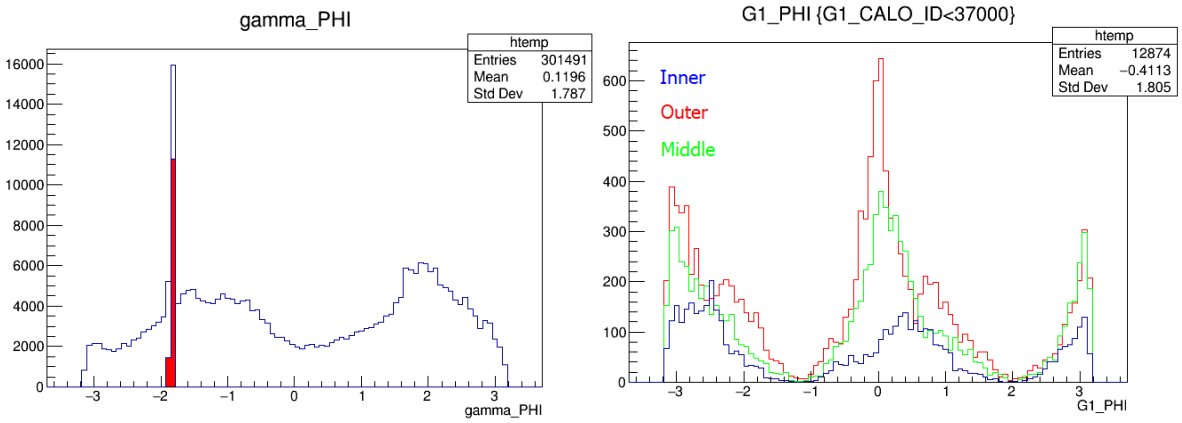
Table 5: The offline selection used in 2022 and 2023 Hlt2 exclusive line triggered candidates. The $\epsilon_{\rm rec}$ is estimated from the upgrade MC (VELO closed, UT included, with nu=7.6), including the tracking/PID/clustering efficiency, Hlt2 selection efficiency, and the offline selection efficiency, the geometry acceptance and generation level cut efficiency are not included.

the distributions of the variables without cuts are shown in Fig. **??** with 2022 data and upgrade MC samples (VELO closed, UT included, nu=7.6). The cut points in most of the distributions are efficient and reasonable.

The double peak distributions of `IsNotH` are quite different from those in Run1 and Run2 data and MC, they are from the updated PID model training strategy. In Run1 and Run2, the model was trained with photons from all three ECAL regions, using variables including CaloHypo_ClusterMatch, CaloHypo_ToPrsE, CaloHypo_E19, CaloHypo_Hcal2Ecal, CaloHypo_PrsE19, CaloHypo_PrsE49, CaloHypo_Spread, CaloHypo_PrsE4Max, CaloHypo_HypoPrsM and CaloHypo_SpdM, while in Run3, 3 independent models are trained separately for inner, middle and outer ECAL region, using the energy fraction of each cell in the 3x3 cluster as input variables. Due to the large cell size in the outer ECAL region (121.9 mm), the neural network model performance is not as good as the models trained on the inner and middle ECAL region photons, so for the outer region photon, the IsNotH output is peak around 0.9. Considering the performance difference in IsNotH, if a tighter photon PID cut is used in selections, like IsNotH $> 0.8$ in Run1 and Run2, then choosing the cut point separately for the outer and the other ECAL regions should be considered.

An abnormal bump was also found in the $\phi$ distribution of the photon in 2023 data, after checking with the cell ID, it is found that this bump is from one certain ECAL cell in the inner region (cell 9629), shown in Fig. 1. In addition, the asymmetry in $\phi$ angle distribution has been found in the inner ECAL region photons, this comes from the partially-closed VELO in 2023 data-taking time, which leads to the worse track reconstruction performance for the high momentum particles ($\eta > 4$), including photons because the cluster is reconstructed with the requirement that no high momentum tracks around.

Another problem that has been found in the check is the missing Hcal energy of photons, this bug was fixed with `Rec v35r18` (related to `Moore v54r19`).




(a) $\phi$ angle distribution of the photons

(b) $\phi$ angle distribution of the photons (noisy cell removed)

Figure 1: The $\phi$ distribution of the photons from Hlt2 exclusive line (BdToKstGamma) triggered 2023 data.In (a), events in the red histogram are from the noisy cell 9629, and this cell has been masked in the later data-taking in 2023. (b) The $\phi$ angle distribution is split into different ECAL regions(inner, middle, outer). The distribution from inner region photons is not as symmetric as photons from the other two regions.

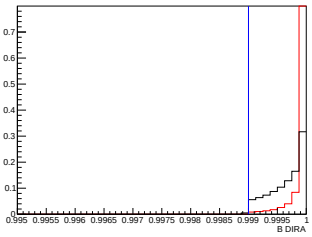
## 2.1 Checks in mass distribution

After checking the basic variables of tracks, photons, and the B meson candidates, the mass distributions of the $\phi$, $K^*$ and $B$ candidates are checked (without the mass window cuts on them).

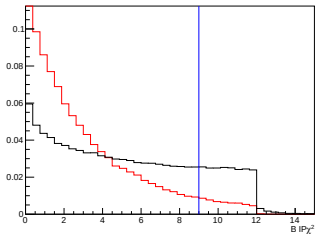
A bump structure in $K^+K^-$ invariant mass distribution near the KK mass threshold has been found in 2022 data, which come from the process of photons converting into electron-positron pairs and the leptons misidentified as kaons. This has been confirmed with the open angle distribution of the two tracks (close to 0, indicating they are sharing the same VELO hits), the vertex position of the two tracks (it's formed like a ring in the X:Y 2D scattering plots), and the invariant mass distributions of the two tracks with electron mass hypothesis (the mass peak is close to 0, as expected since the photon conversion process is not a physics decay and there's energy and momentum loss in the electron-positron pair system), shown in Fig. 10. This bump can be easily removed by the cut on the open angle ($> 0.0004$ for 2022 and $> 0.0015$ for 2023).

With the open angle cut on the two tracks, the fits to $\phi$ and $K^*$ mass have been done to check both the mass peak positions and the resolutions. Assuming there are only signal and combinatorial background in the data. Two Breit-Wigner functions are used to describe the signals, the widths of the signals are fixed to the PDG value for both $K^*$ and $\phi$, and the mean value of the peak is set to be float. To describe the resolution effect, the Gaussian function is convoluted to the signal Breit-Wigner function, the mean value of this resolution function is set to be 0 while the width is set to be float. The fit results of two channels in two years are shown in Tab. 6 and Fig. 11.

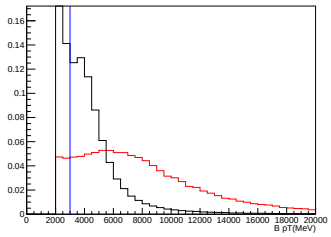
There is no clear signal peak has been observed in the Hlt2 exclusive line triggered data, both for 2022 and 2023. However, after a simple pT cut on the photon (pT¿2.6 GeV), we do find quite significant signal peaks of $B_0$ and $B_s$ in 2023 spruce line triggered data. A Gaussian function is used to describe the signal, with mean and width being set



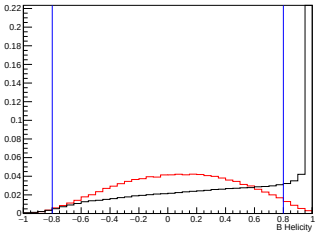

(a) $B^0 \to K^*\gamma$ B DIRA



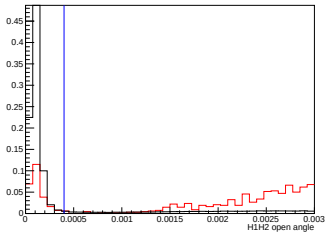

(b) $B^0 \to K^*\gamma$ B IPCHI2



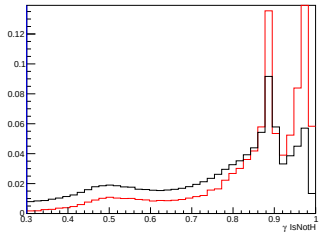

(c) $B^0 \to K^*\gamma$ B pT



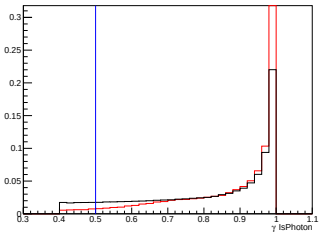

(d) $B^0 \to K^*\gamma$ B helicity



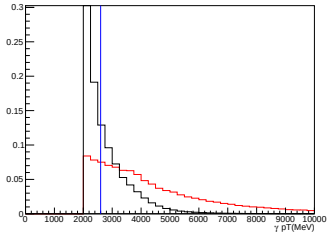

(e) $B^0 \to K^*\gamma$ Open angle



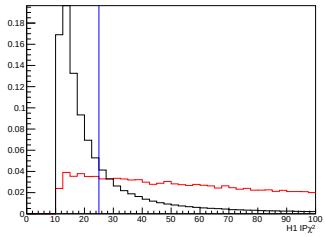

(f) $B^0 \to K^*\gamma$ photon IsNotH



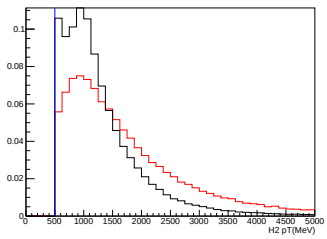

(g) $B^0 \to K^*\gamma$ photon IsPhoton



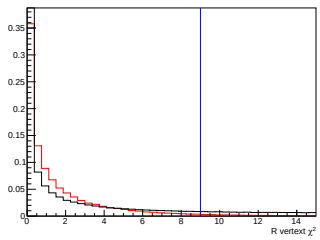

(h) $B^0 \to K^*\gamma$ photon pT




(i) $B^0 \to K^*\gamma$ Track IPCHI2

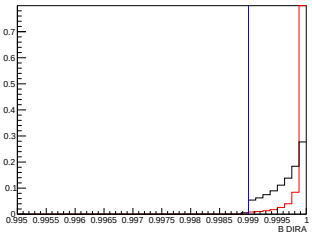



(j) $B^0 \to K^*\gamma$ Track pT

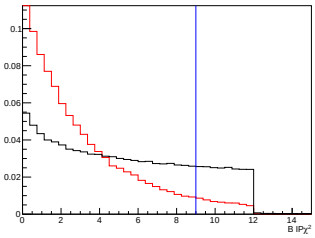



(k) $B^0 \to K^*\gamma$ $K^*$ vertex chi2

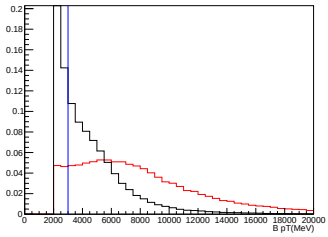
Figure 2: Comparison between upgrade MC and 2022 Hlt2 exclusive lines triggered data, the red histograms are from signal MC, black histograms are from 2022 data and the blue line is the cut point of the selections.



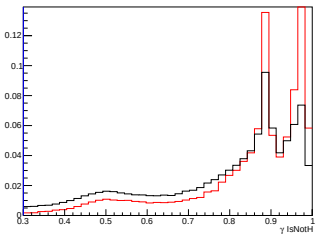

(a) $B^0 \to K^*\gamma$ B DIRA



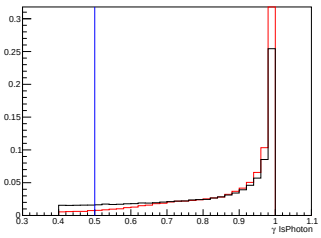

(b) $B^0 \to K^*\gamma$ B IPCHI2



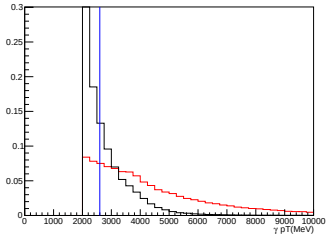

(c) $B^0 \to K^*\gamma$ B pT



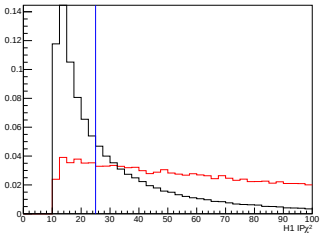

(d) $B^0 \to K^*\gamma$ photon IsNotH



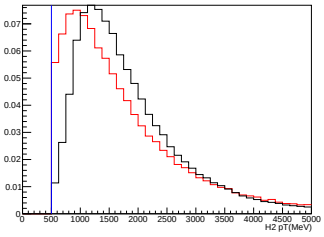

(e) $B^0 \to K^*\gamma$ photon IsPhoton



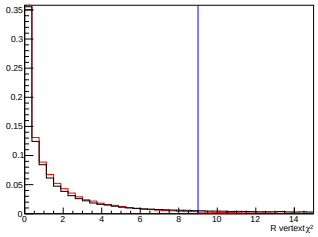

(f) $B^0 \to K^*\gamma$ photon pT




(g) $B^0 \to K^*\gamma$ Track IPCHI2




(h) $B^0 \to K^*\gamma$ Track pT

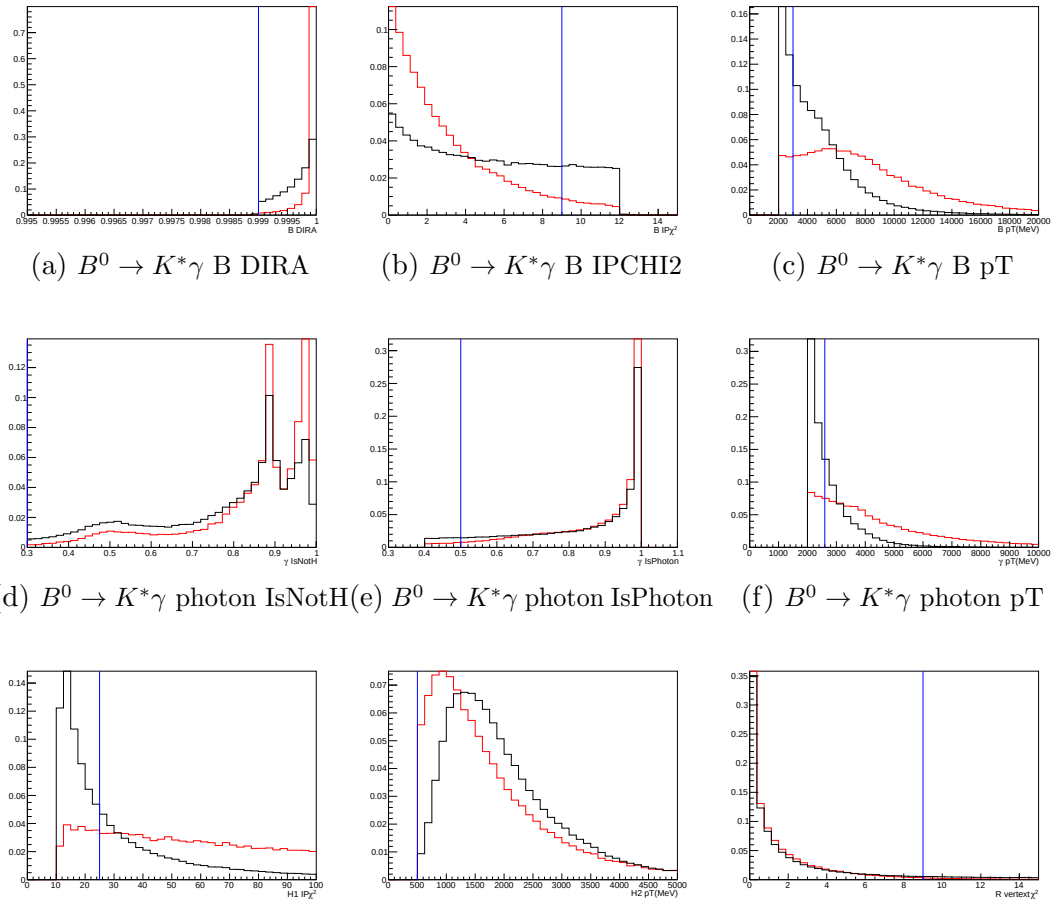



(i) $B^0 \to K^*\gamma$ $K^*$ vertex chi2

Figure 3: Comparison between upgrade MC and 2023 Hlt2 exclusive lines triggered data (commissioning23), the red histograms are from signal MC, black histograms are from 2022 data and the blue line is the cut point of the selections.

| | MC | 2022 Data | 2023 Commission | 2023 Collision | 2023 Spruce |
|---|---|---|---|---|---|
| $\mu_\phi$ | $1019.7 \pm 0.5$ | $1019.1 \pm 0.2$ | $1019.6 \pm 0.02$ | $1019.5 \pm 0.02$ | $1019.4 \pm 0.14$ |
| $\sigma_\phi$ | $1.23 \pm 0.83$ | $3.16 \pm 0.19$ | $1.11 \pm 0.03$ | $1.05 \pm 0.04$ | $0.83 \pm 0.34$ |
| $\mu_K$ | $897.1 \pm 0.1$ | $898.2 \pm 1.2$ | $894.6 \pm 0.3$ | $894.8 \pm 0.3$ | $894.2 \pm 1.4$ |
| $\sigma_K$ | $3.20 \pm 0.78$ | $7.42 \pm 2.01$ | $3.94 \pm 1.10$ | $1.89 \pm 2.06$ | $2.17 \pm 5.3$ |

Table 6: $K^*$ and $\phi$ mass fit result in different data samples.




(a) $B^0 \to K^*\gamma$ B DIRA (b) $B^0 \to K^*\gamma$ B IPCHI2 (c) $B^0 \to K^*\gamma$ B pT

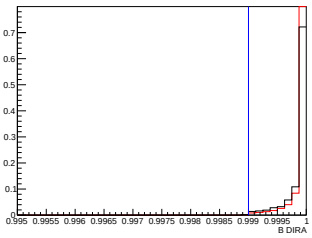
(d) $B^0 \to K^*\gamma$ photon IsNotH (e) $B^0 \to K^*\gamma$ photon IsPhoton (f) $B^0 \to K^*\gamma$ photon pT

(g) $B^0 \to K^*\gamma$ Track IPCHI2 (h) $B^0 \to K^*\gamma$ Track pT (i) $B^0 \to K^*\gamma$ $K^*$ vertex chi2

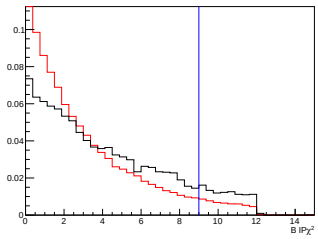
Figure 4: Comparison between upgrade MC and 2023 Hlt2 exclusive lines triggered data (collision23), the red histograms are from signal MC, black histograms are from 2022 data and the blue line is the cut point of the selections.

to float, and an exponential function is employed to simply describe the background since there are no 2023 background MC samples for background study in detail. The fit results are shown in Fig. 12. To validate the existence of signals, different cuts are performed to check if the signal vanished with the changes of selections.

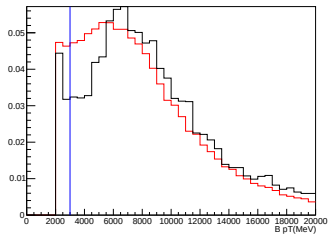
It has been noticed that when having a similar selection on $B_s$ and $B_0$ channels, similar signal yields are seen in the data. However, based on the Run1 and Run2 measurements, there should be around 6 times more signal yields in $B_0$ than in $B_s$. After the comparison of the spruce lines in these two channels, a difference in DIRA cut is found in the code. For the $B_s \to \phi\gamma$ channel, the DIRA is required to be larger than cos(0.1) while for $B^0 \to K^*\gamma$ this requirement is set to be cos(0.045) for suppressing the trigger rate. To evaluate the effect of this cut difference, two fits are down on the $B_s \to \phi\gamma$ channel with 2023 spruce line triggered data, with and without the cos(0.045) cut. The efficiency loss of this cut is found to be 16.5%, which is probably not the only factor that lowers the $B^0 \to K^*\gamma$ signal yields. Furthermore, the mass window cut on $\phi$ with spruce line output also changes the $B^0$ to $B_s$ ratio, after performing the mass window ($\pm$ 9 MeV), and the



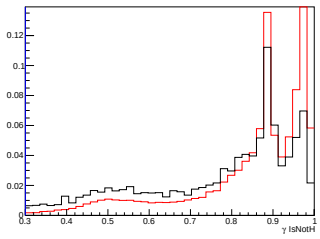

(a) $B^0 \to K^*\gamma$ B DIRA



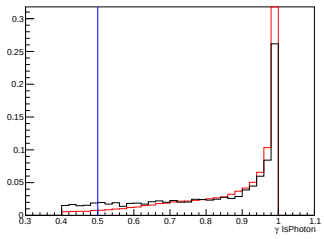

(b) $B^0 \to K^*\gamma$ B IPCHI2



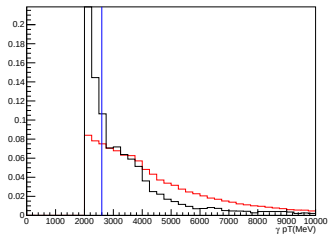

(c) $B^0 \to K^*\gamma$ B pT



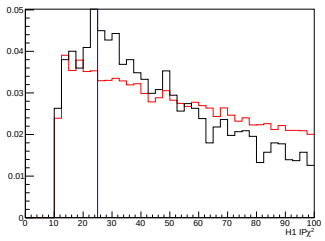

(d) $B^0 \to K^*\gamma$ photon IsNotH



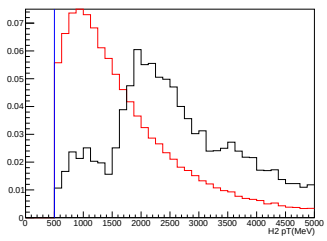

(e) $B^0 \to K^*\gamma$ photon IsPhoton



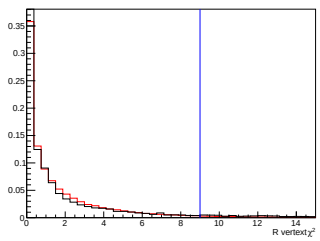

(f) $B^0 \to K^*\gamma$ photon pT




(g) $B^0 \to K^*\gamma$ Track IPCHI2



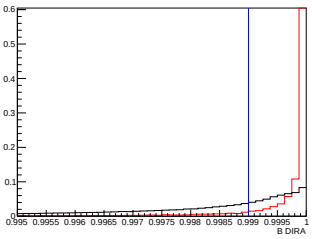

(h) $B^0 \to K^*\gamma$ Track pT



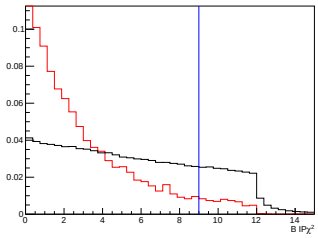

(i) $B^0 \to K^*\gamma$ $K^*$ vertex chi2

Figure 5: Comparison between upgrade MC and 2023 Spruce exclusive lines triggered data (collision23), the red histograms are from signal MC, black histograms are from 2022 data and the blue line is the cut point of the selections. To be clear, the cuts are actually not performed on the spruce data, just for comparison with Hlt2 exclusive line triggered MC and data.

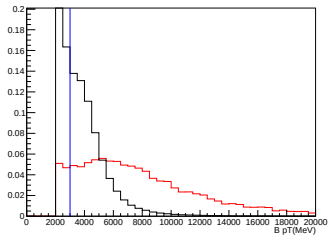
ratio gets larger with 35%. These two factors are expected to change the ratio from 1.3 to 2.3, estimated from data directly. Further checks need to be done with the 2023 condition signal MC and background MC.



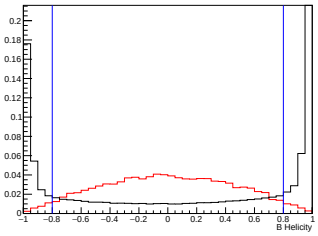

(a) $B_s \to \phi\gamma$ B DIRA



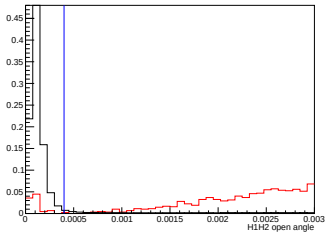

(b) $B_s \to \phi\gamma$ B IPCHI2



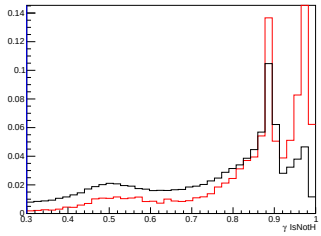

(c) $B_s \to \phi\gamma$ B pT



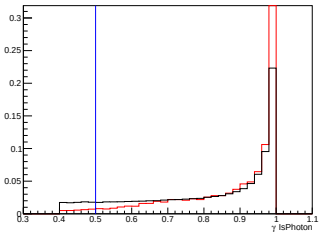

(d) $B_s \to \phi\gamma$ B helicity



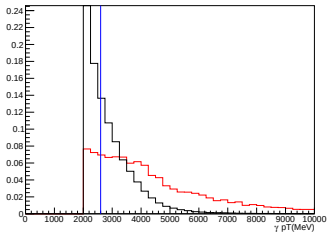

(e) $B_s \to \phi\gamma$ Open angle



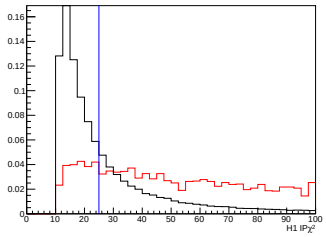

(f) $B_s \to \phi\gamma$ photon IsNotH



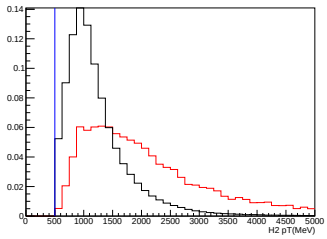

(g) $B_s \to \phi\gamma$ photon IsPhoton



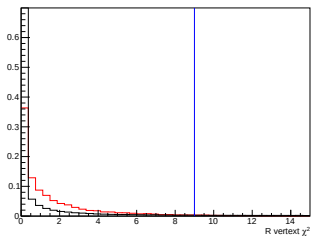

(h) $B_s \to \phi\gamma$ photon pT




(i) $B_s \to \phi\gamma$ Track IPCHI2

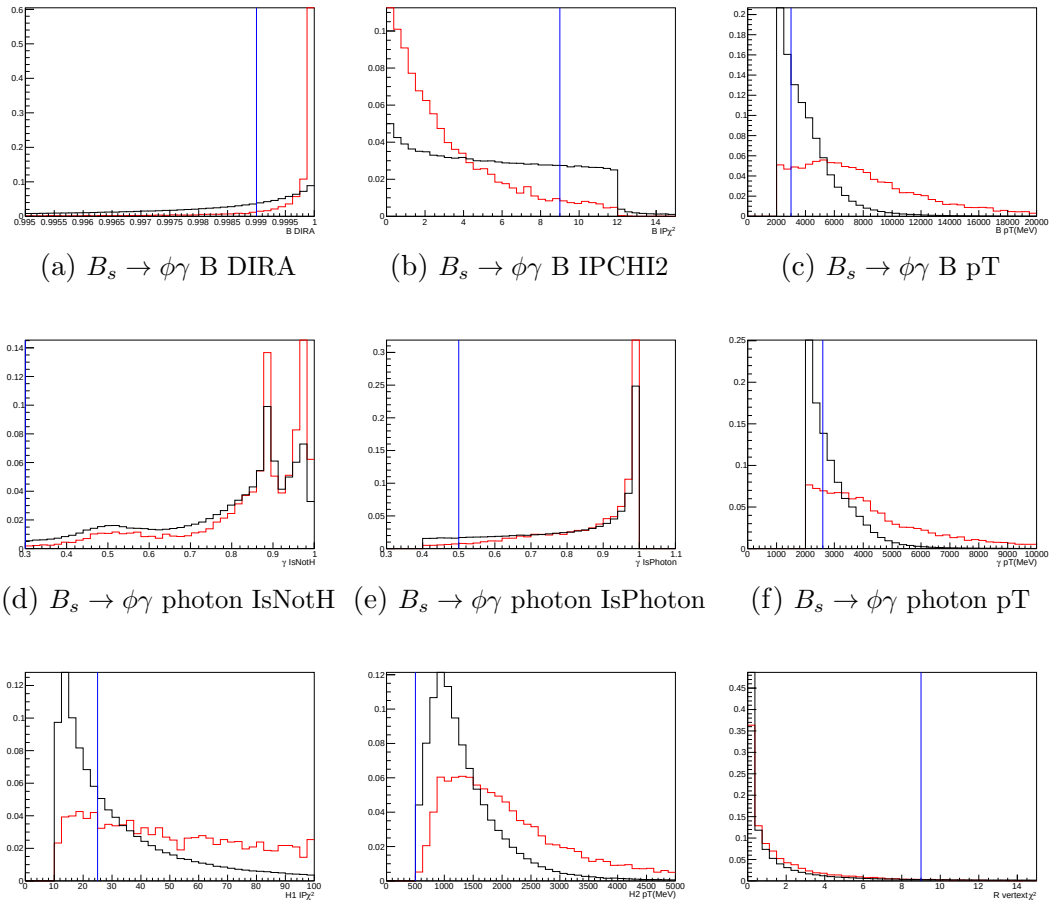



(j) $B_s \to \phi\gamma$ Track pT




(k) $B_s \to \phi\gamma$ $K^*$ vertex chi2

Figure 6: Comparison between upgrade MC and 2022 Hlt2 exclusive lines triggered data, the red histograms are from signal MC, black histograms are from 2022 data and the blue line is the cut point of the selections.

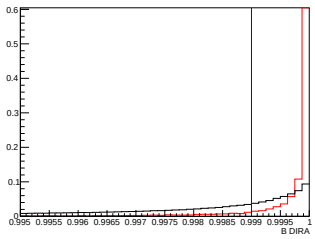



(a) $B_s \to \phi\gamma$ B DIRA (b) $B_s \to \phi\gamma$ B IPCHI2 (c) $B_s \to \phi\gamma$ B pT

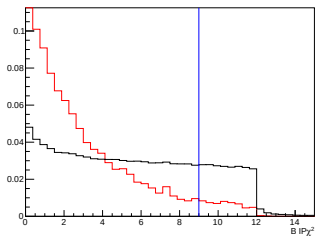
(d) $B_s \to \phi\gamma$ photon IsNotH (e) $B_s \to \phi\gamma$ photon IsPhoton (f) $B_s \to \phi\gamma$ photon pT

(g) $B_s \to \phi\gamma$ Track IPCHI2 (h) $B_s \to \phi\gamma$ Track pT (i) $B_s \to \phi\gamma$ $K^*$ vertex chi2

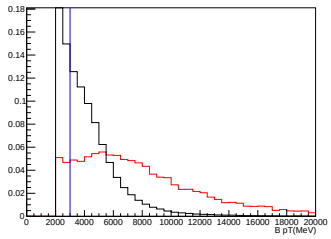
Figure 7: Comparison between upgrade MC and 2023 Hlt2 exclusive lines triggered data (commissioning23), the red histograms are from signal MC, black histograms are from 2022 data and the blue line is the cut point of the selections.



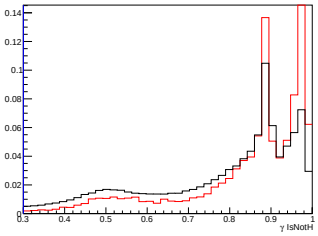

(a) $B_s \to \phi\gamma$ B DIRA



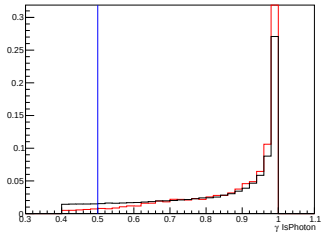

(b) $B_s \to \phi\gamma$ B IPCHI2



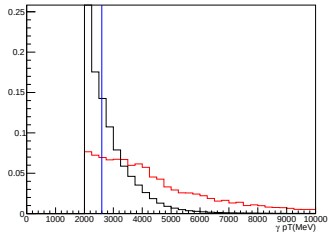

(c) $B_s \to \phi\gamma$ B pT



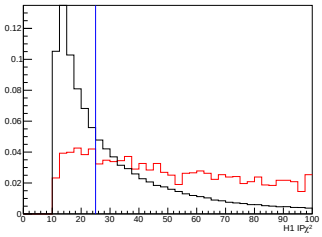

(d) $B_s \to \phi\gamma$ photon IsNotH



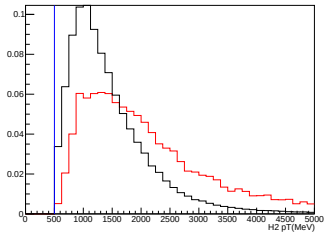

(e) $B_s \to \phi\gamma$ photon IsPhoton



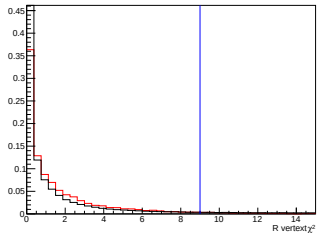

(f) $B_s \to \phi\gamma$ photon pT




(g) $B_s \to \phi\gamma$ Track IPCHI2

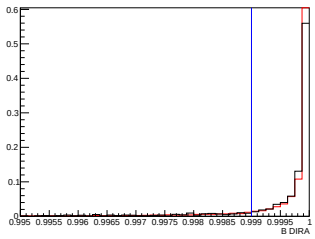



(h) $B_s \to \phi\gamma$ Track pT

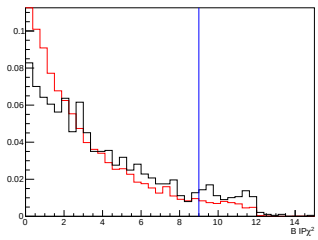



(i) $B_s \to \phi\gamma$ $K^*$ vertex chi2

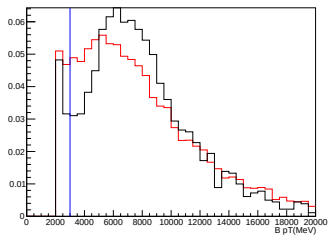
Figure 8: Comparison between upgrade MC and 2023 Hlt2 exclusive lines triggered data (collision23), the red histograms are from signal MC, black histograms are from 2022 data and the blue line is the cut point of the selections.



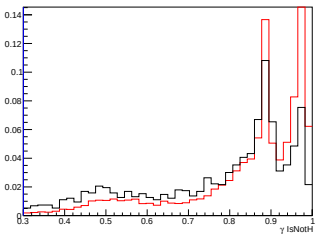

(a) $B_s \to \phi\gamma$ B DIRA



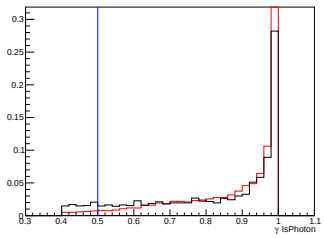

(b) $B_s \to \phi\gamma$ B IPCHI2



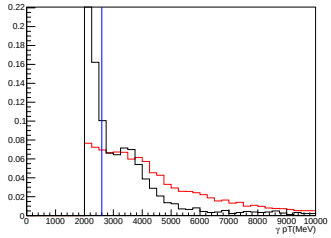

(c) $B_s \to \phi\gamma$ B pT



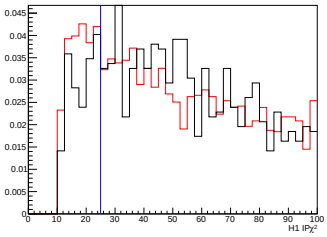

(d) $B_s \to \phi\gamma$ photon IsNotH



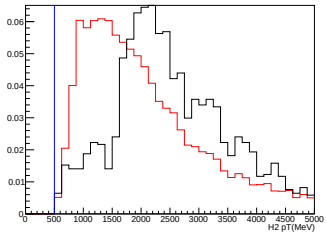

(e) $B_s \to \phi\gamma$ photon IsPhoton



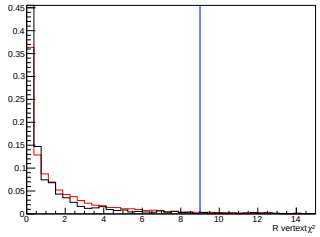

(f) $B_s \to \phi\gamma$ photon pT




(g) $B_s \to \phi\gamma$ Track IPCHI2




(h) $B_s \to \phi\gamma$ Track pT




(i) $B_s \to \phi\gamma$ $K^*$ vertex chi2

Figure 9: Comparison between upgrade MC and 2023 Spruce lines triggered data (collision23), the red histograms are from signal MC, black histograms are from 2022 data and the blue line is the cut point of the selections. To be clear, the cuts are not performed on the spruce data.

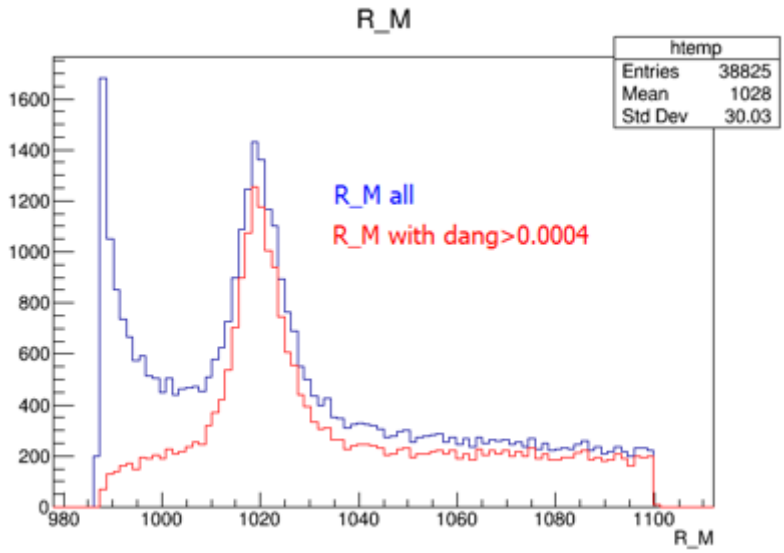
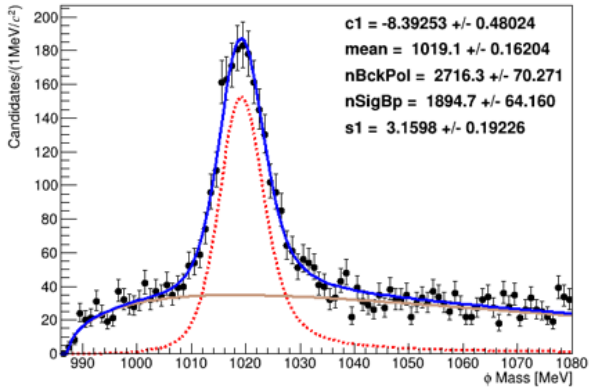
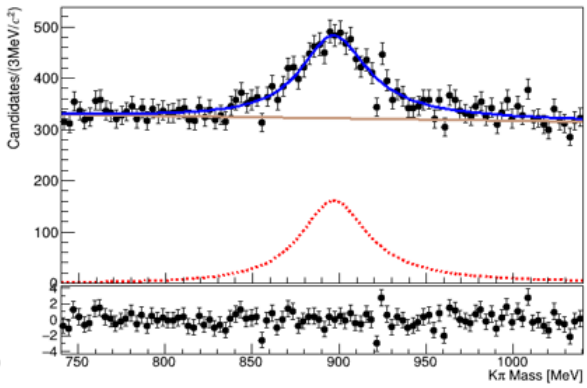


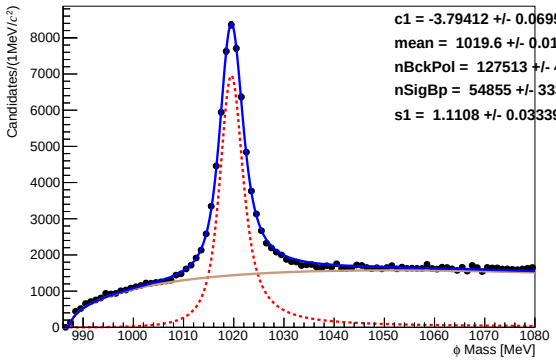

Figure 10: The potential photon conversion events form aa bump at KK mass threshold. The red histogram shows the KK mass distribution after the open angle cut, the bump is removed by this cut completely.



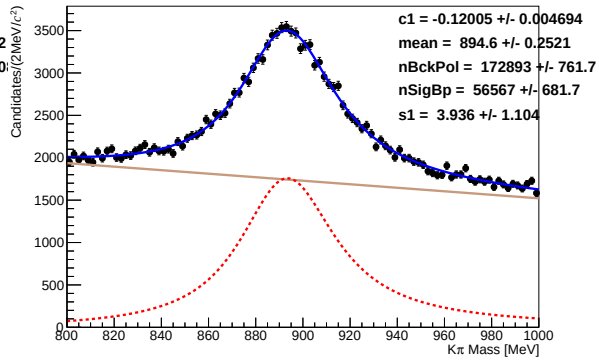

(a) $B_s \to \phi\gamma$ 2022 data



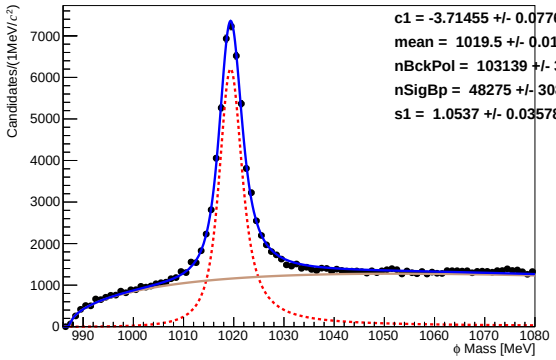

(b) $B^0 \to K^*\gamma$ 2022 data



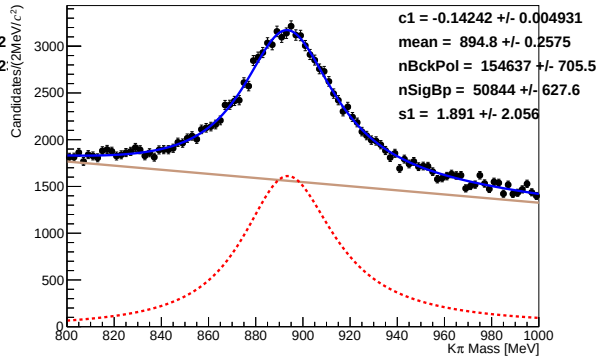

(c) $B_s \to \phi\gamma$ Commissioning23 data



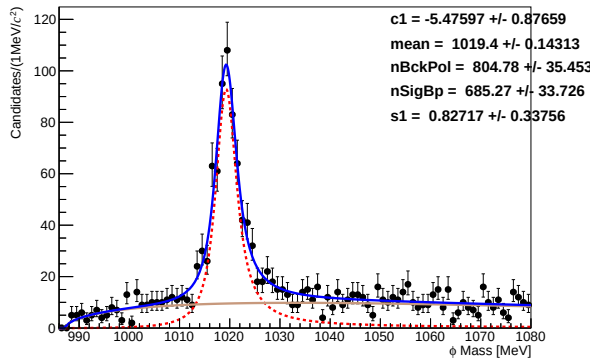

(d) $B^0 \to K^*\gamma$ Commissioning23 data



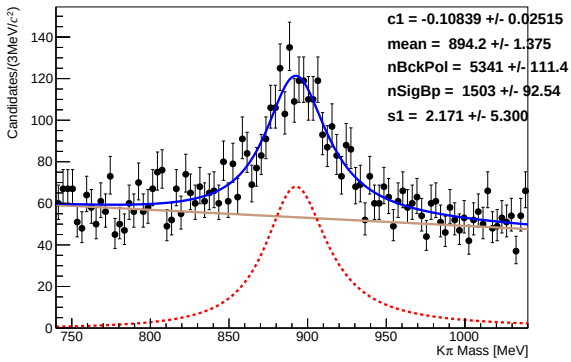

(e) $B_s \to \phi\gamma$ Collision23 data




(f) $B^0 \to K^*\gamma$ Collision23 data



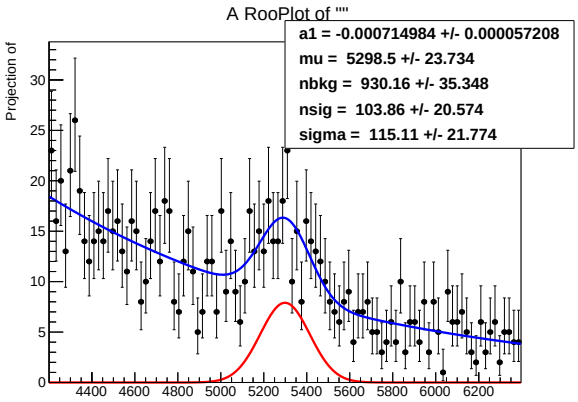

(g) $B_s \to \phi\gamma$ Full stream spruce23 data



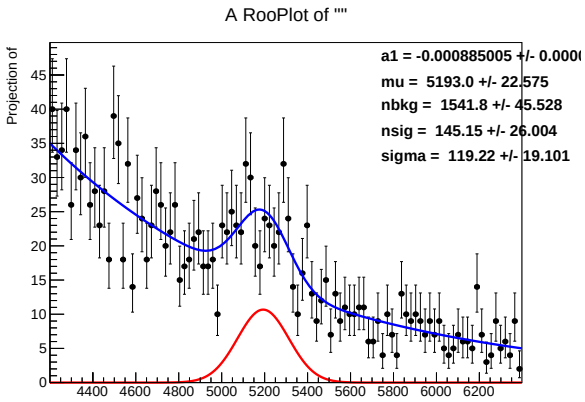

(h) $B^0 \to K^*\gamma$ Full stream spruce23 data



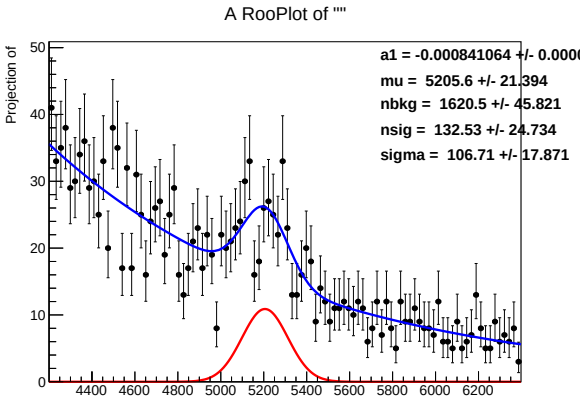
Figure 11: Fit plots of the $K\pi$ and $KK$ mass distributions in different data samples.



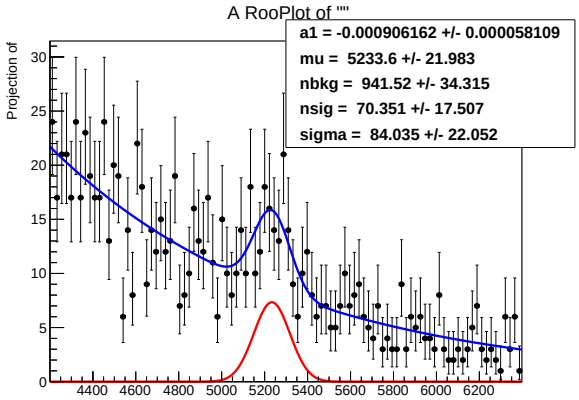

(a) $B_s \to \phi\gamma$ 2023 data, photon pT¿2.6 GeV and K_PID




(b) $B^0 \to K^*\gamma$ 2023 data, photon pT¿2.6 GeV and K_PID

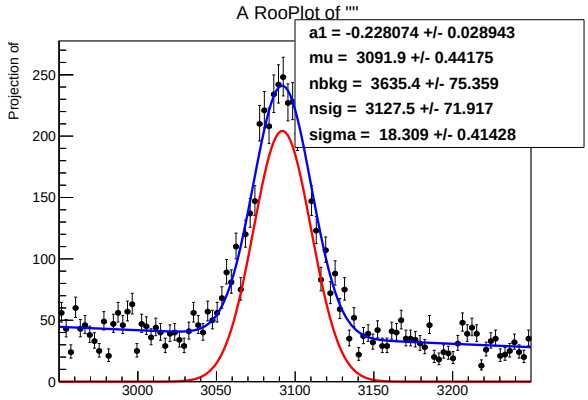



(c) $B^0 \to K^*\gamma$ 2023 data, photon pT¿2.6 GeV and Kst mass window (45 MeV)

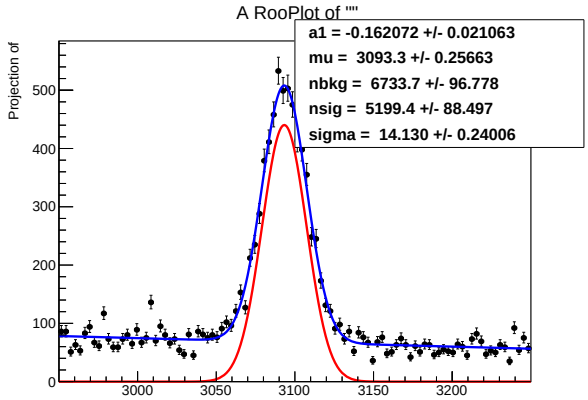



(d) $B^0 \to K^*\gamma$ 2023 data, photon pT¿2.6 GeV, only in outer region, Kst mass window (45 MeV),

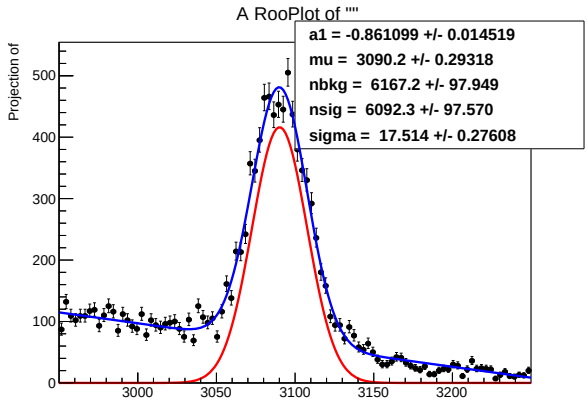
Figure 12: Fit plot of $B_0$ and $B_s$ mass in 2023 spruce line triggered data



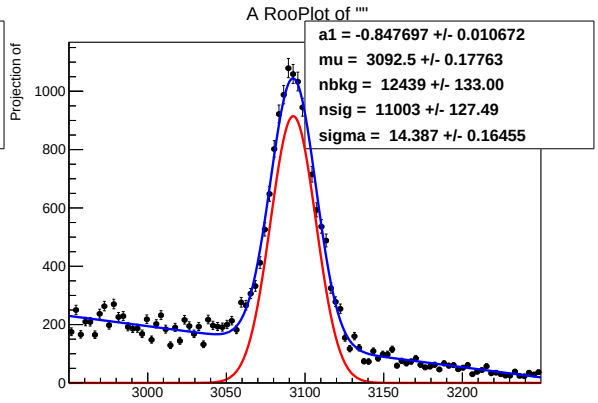

(a) BuToJpsiKstPlus, Commission23




(b) BuToJpsiKstPlus, Collision23




(c) BuToChicKPlus, Commision23




(d) BuToChicKPlus, Coolision23

Figure 13: Fit to $J/\psi$ in 2023 commissioning data and 2023 collision data

# 3 Charmonia channels

After the basic selection, listed in Tab. 3, simple fits to $J/\psi$ are done on the two charmonia channels, shown in Fig. 13. In this simple fit, a Gaussian function is employed as the signal shape of $J/\psi$, the mean and sigma values float, and a Chebychev function is used to describe the background. The fit results are shown as well on the fit plots. The resolutions of $J/\psi$ are consistent within two charmonia channels in commissionning23 and collision23 data, and there is a quite clear resolution improvement in collision23 data, compared with the commissioning23 data. There is a slight shift in the mean values of the $J/\psi$ mass peak, comparing data with the PDG value (3096.9 MeV), this shift is improved in collision23 data, but still there. This could come from the imperfect alignment and calibration version that used in the data processing.

Despite the checks with $J/\psi$, the checks with photon/$\pi^0$ are done as well, with the tracks ($J/\psi$ for $\chi_c$, $K$ for $K^*$). For the

# 4 Conclusion and next-to-do

# References

[1] LHCb collaboration, R. Aaij *et al.*, *The LHCb Upgrade I*, `arXiv:2305.10515`, submitted to JINST.